StudyForge — Complete Product Document

What We Are Building

Product Name

StudyForge — AI-Powered Study Material Transformer

One Line Description

An app that takes any study material (file or topic) and transforms it into comprehensive, exam-ready or learning-ready notes with intelligent visuals, current affairs, previous year questions, and subject-adaptive features.

Who Is It For

- Students preparing for UPSC, State PSC, or any competitive exam
- Students studying academic subjects like Computer Science, Literature, Medicine
- Anyone who wants to deeply understand a topic with visual explanations
- Self-learners who want organized notes from raw material

How It Works (Simple)

User gives input → App reads and understands → App plans how to teach → App writes notes with images and extras → User reviews and approves → User downloads final document

The Three Starting Points

Starting Point A: User Has A File

User uploads study material they already have. Supported file types:
- PDF
- TXT (plain text / notepad files)
- DOC / DOCX (Word documents)
- HTML (web pages saved as files)
- MD (Markdown files)
- Images (JPG, PNG — text extracted from images)

Starting Point B: User Has A Topic Name

User doesn't have a file. They just type:
- Main topic name (required)
- Sub-topics (optional, can add multiple)
- Specific focus area (optional)

App generates comprehensive raw content on that topic and treats it as if it were an uploaded document.

Starting Point C: User Pastes Text

User copies text from somewhere and pastes it directly into the app.

What The App Produces

The Final Output Contains

- Organized notes broken into logical parts
- Visual explanations — diagrams, charts, comics, flowcharts, tables, infographics
- Current affairs relevant to the topic (if applicable)
- Previous year questions detected from source + analysis
- Comparison tables where concepts need side-by-side analysis
- Timelines where historical progression matters
- Legal/case law boxes where constitutional or legal aspects exist
- Stories and scenarios to make concepts memorable
- Code examples and output previews (for programming subjects)
- Input/Output demonstrations (for technical tools and algorithms)

Output Formats

- HTML (styled webpage)
- PDF (via browser print)
- Markdown (plain text with formatting)
- Copy to clipboard

The 8 Screens

1. **Screen 1: INPUT** — User provides study material and preferences
2. **Screen 2: READING** — App processes the input (progress display)
3. **Screen 3: UNDERSTANDING + FEATURE DETECTION** — App shows what it understood and which features apply
4. **Screen 4: PARTS** — App proposes how to split material into teachable parts
5. **Screen 5: BLUEPRINT (Card Grid)** — App shows high-level teaching plan for each part
6. **Screen 6: BLUEPRINT DETAIL VIEW** — Interactive workspace to chat with AI and edit exact prompts/plans for a specific part
7. **Screen 7: GENERATION (Single Long Page)** — App writes notes, generates images, fetches CA — user reviews
8. **Screen 8: EXPORT** — User downloads final compiled document

Subject-Adaptive Behavior

The app is NOT limited to one subject. It adapts based on what the content is about.

For Exam Preparation (UPSC / PSC / Competitive)

Feature	Visible	Purpose
Current Affairs	✅ Yes	Latest news related to topics
State Focus (MP, UP, etc.)	✅ Yes	State-specific data and schemes
PYQ Detection	✅ Yes	Find and tag previous year questions

Legal/Case Law	✅ If applicable	Constitutional provisions, court cases
Timeline	✅ If historical	Evolution of policies, movements
Comparison Tables	✅ Yes	GDP vs GNP, Fundamental Rights vs DPSP
Formula Breakdown	✅ If applicable	Economics formulas, science formulas
Story/Scenario Boxes	✅ Yes	Memorable examples with characters

For Computer Science / Programming

Feature	Visible	Purpose
Code Blocks	✅ Yes	Syntax-highlighted code examples
Algorithm Tables	✅ Yes	Time/space complexity comparison
Input → Output Demos	✅ Yes	Show what tool/function produces
Architecture Diagrams	✅ Yes	System design, data flow
Data Structure Visuals	✅ Yes	Trees, graphs, stacks, queues
Current Affairs	❌ Hidden	Not needed (can force-enable)
State Focus	❌ Hidden	Not relevant (can force-enable)
PYQ Detection	⚠️ Maybe	If academic exam questions exist

For Literature / Humanities

Feature	Visible	Purpose
Timeline	✅ Yes	Literary periods, author lifetimes
Character Maps	✅ Yes	Relationships between characters
Theme Comparisons	✅ Yes	Themes across works
Current Affairs	❌ Hidden	Not needed (can force-enable)
Formulas	❌ Hidden	Not relevant
Code Blocks	❌ Hidden	Not relevant

For Medical / Science

Feature	Visible	Purpose

Diagrams	✅ Yes	Anatomy, processes, cycles
Formulas	✅ Yes	Chemical equations, physics formulas
Comparison Tables	✅ Yes	Drug comparisons, disease vs disease
Timeline	✅ If applicable	Disease progression, treatment evolution
Current Affairs	⚠️ Maybe	Medical breakthroughs, policy changes

Feature Detection Logic

After the app reads the content (Screen 3), it automatically detects which features are relevant based on:

- Subject matter identified
- Content types found (formulas, dates, code, legal references)
- User's stated exam/objective
- What elements were extracted (PYQs, tables, diagrams)

User can always override — turn OFF detected features or turn ON hidden ones via "Show more options" link.

Image Generation Philosophy

Not One Style — Mixed Styles

A single image can combine multiple visual styles when the content demands it.
Example: GDP vs GNP image could contain:

- Flowchart showing where GDP and GNP sit in economics hierarchy
- Bar chart comparing India's GDP vs GNP over years
- Table showing data comparison for India and MP
- Small explanation cards on the side describing behavior patterns

Styles Available (35 Options Grouped)

- **Process/Flow:** Flowchart, Step-by-Step, Decision Tree, Sankey, Process Map, Timeline, Gantt
- **Structure:** Entity Diagram, Hierarchical Tree, Network, Sequence Diagram, Use Case, State Machine, Mind Map
- **Data:** Bar Chart, Pie Chart, Line Graph, Scatter Plot, Heat Map, Radar, Bubble Chart, Stacked Area, Pictograph, Heat-Map Table
- **Concept:** Concept Map, Venn Diagram, Comparison Matrix, Comic Story, Infographic, 3D Model, Icon Array, Geographic Map, Funnel, Storyboard, Table Diagram

How Images Are Created

- AI automatically picks the best style(s) based on content
- User can change to any style via dropdown
- AI generates a detailed prompt for image creation
- User can edit the prompt before generation
- User can regenerate with different style
- User can describe their own vision and AI builds the prompt

Interactive Image Creation

User highlights any text in notes → 🎨 button appears → shows editable auto-generated prompt → shows recommended styles → user picks/modifies → image generates → user approves or regenerates

AI Validation System

At every stage, the app can validate its own work:

What AI Validation Checks

- Text authenticity — are facts correct and traceable
- Image relevance — does the image actually explain the concept
- Coverage completeness — are all planned topics actually covered
- Suggestions — how to improve text, images, or structure
- Missing elements — anything that should be added

Where Validation Happens

- Blueprint stage: AI reviews the teaching plan before generation
- Generation stage: AI reviews generated notes per part
- Image stage: AI reviews each image and suggests improvements
- User-triggered: User clicks "AI Validate" button anytime

Error Handling — Universal

Everywhere in the app where AI is called, the same error handling applies:

1. **Step 1**: AI call is made
2. **Step 2**: If it fails → auto-retry (wait 2 seconds)
3. **Step 3**: If it fails again → auto-retry (wait 4 seconds)
4. **Step 4**: If it still fails → show popup:

```
❌ Failed to [action name]        │
                        │
Reason: [error details]        │
                        │
[🔄 Try Again]              │
[✏️ Modify Prompt & Try Again]     │
[⏭️ Skip This Step]            │
[❌ Cancel]                 │
```

This applies to:

- Text generation
- Image generation
- Current affairs search
- Blueprint generation
- AI validation
- AI chat responses
- Any AI interaction anywhere

Approval System

Three Levels Of Approval

Level 1: Individual Element
- Approve specific text section
- Approve specific image
- Approve specific CA integration

Level 2: Part Level
- "Approve Part" button — approves all elements in that part
- Locks the part with green checkmark
- Shows "Unlock to Edit" button

Level 3: All Parts
- "Approve All" master button
- Only available when all parts have been reviewed

Progress Tracking

- Top progress bar: "3 of 6 parts approved"
- Sidebar: Each part has checkmark/pending/generating icon
- Floating widget: Percentage complete (if not too heavy on performance)

Unlock to Edit

After approving a part, user can click "Unlock to Edit" to:
- Re-open the part for modifications
- Make changes
- Re-approve when done
- Progress indicator updates accordingly

Traceability System

Every piece of content in the final notes is tagged:
- 📗 Source-Backed (Green): Directly from uploaded material — verified content
- 🔍 Research-Required (Blue): Content that was researched/generated by AI — should be verified
- 💎 Optional Enrichment (Gold): Extra content added for better understanding — nice to have

This helps the user know what's from their original material and what AI added.

Save and Resume

- App automatically saves all progress to browser storage
- If user closes browser and comes back, everything is exactly where they left
- No manual save button needed
- Works across page refreshes
- State includes: all inputs, all AI outputs, all approvals, all generated content

Navigation

Methods Available

- Sidebar: Shows all stages/screens, click to jump to any completed stage
- Top progress bar: Visual indicator of current stage with clickable stages
- Back button: Go to previous screen
- Forward button: Only available if next stage is already completed

Rules

- Can always go back to any completed screen
- Can only go forward if current screen requirements are met
- Blueprint approval required before generation starts
- Part generation required before export

Prompt Architecture (Planning — Not Building Yet)

The app works through a chain of AI prompts where each prompt's output feeds the next:

Plaintext
```
Stage 1 Output → Feeds → Blueprint Generation Prompt
Blueprint Prompt Output → Feeds → Per-Part Planning Prompts
Per-Part Plan → Feeds → Text Generation Prompt
Per-Part Plan → Feeds → Image Generation Prompt
Per-Part Plan → Feeds → CA Search Prompt
Generated Content → Feeds → Validation Prompt
User Selection → Feeds → Visualize-This Prompt
```

Key principle: Every prompt is visible, copyable, editable, and regenerable. User can see exactly what the AI is being asked, modify it, and regenerate. We will plan the detailed prompt chain when we reach the implementation phase.

Non-Functional Requirements

Performance

- App should work smoothly with documents up to 500 pages
- Single page app — no page reloads
- Progress indicators for any operation taking more than 2 seconds
- Auto-save should not cause UI lag

Reliability

- Auto-retry on AI failures (2 retries with delay)
- User-friendly error messages (not technical errors)
- No data loss on browser refresh
- Graceful handling of unsupported file types

Usability

- Minimal, moderate design — not fancy but clean
- All interactive elements clearly labeled
- No mandatory fields that block progress unnecessarily
- Placeholder/dummy labels clearly visible during skeleton testing
- Subject-adaptive UI — irrelevant features hidden, not cluttered

Technical Constraints

- Single HTML file (HTML + CSS + JS)
- Uses Gemini's built-in AI (no external API keys)
- No backend server needed
- Works in modern browsers (Chrome, Firefox, Edge)
- All processing happens in browser

Implementation Plan (Overview)

- **Phase 1: Skeleton** - Build the complete UI shell with all 8 screens, navigation, state management, and placeholder content. No real AI logic. Goal: Verify the entire user flow works end-to-end with dummy data.
- **Phase 2: Stage 1 (Real AI)** - Replace placeholders in Screens 1-3 with real file parsing and AI analysis. Goal: Upload a real PDF and see real analysis output.
- **Phase 3: Stage 2 (Real AI)** - Replace placeholders in Screens 4-5 with real partitioning and blueprint generation. Goal: See real blueprint cards with real teaching plans.

- **Phase 4: Stage 3 (Real AI)** - Replace placeholders in Screen 7 (Generation) with real text generation, image generation, CA search, and validation. Goal: See real notes with real images and current affairs.
- **Phase 5: Stage 4 (Export)** - Build real export functionality on Screen 8. Goal: Download a complete, properly formatted document.
- **Phase 6: Polish** - End-to-end testing, bug fixes, performance optimization, edge case handling.

Summary

StudyForge is a single-page AI-powered app that:

- Accepts any study material (file upload, topic name, or pasted text)
- Understands the content (subject detection, topic mapping, element extraction)
- Adapts to the subject (shows relevant features, hides irrelevant ones)
- Plans intelligently (blueprint with text, image, CA, and validation plans)
- Generates comprehensive notes (text + visuals + CA + PYQs + extras)
- Validates its own work (AI checks authenticity and suggests improvements)
- Gives user full control (edit, regenerate, approve at every step)
- Exports polished document (multiple formats)
- All in one HTML file. No external services. No API keys. Just Gemini's built-in AI.

StudyForge — Detailed Stage Specifications

STAGE 1: UNDERSTAND THE SOURCE

Screen 1: Input

What User Sees

Three input methods separated by clean dividers. All settings optional.

Input Method 1: File Upload

- Accepted Types: PDF, TXT, DOC, DOCX, HTML, MD, JPG, PNG
- Multiple Files: No — one file at a time
- File Size Hint: Show file name and size after selection
- Drag & Drop: Nice to have but not mandatory

Input Method 2: Type Topic

- Main Topic: Yes (if using this method) e.g., "National Income"
- Sub-topics: Optional, can add multiple e.g., "GDP", "GNP", "NDP"
- Specific Focus: Optional e.g., "Focus on Indian economy data"
- Add Sub-topic Button: [+ Add another] creates new field. Up to 10 sub-topics.

Input Method 3: Paste Text

- Large Text Area: Resizable, min 10 rows
- Placeholder Text: "Paste your study material here..."
- Character Count: Shows live character count at bottom

Optional Settings (Below All Three Methods)

- Exam: Dropdown (UPSC / State PSC / Academic / Other) Default: Blank
- Objective: Dropdown (Notes / Answer Writing / Revision) Default: Blank
- State Focus: Text Field ("MP" / "UP" / blank) Default: Blank
- Include Current Affairs: Toggle (On / Off) Default: Off

Continue Button

- Always active — user decides when to proceed
- Validates that at least ONE input method has content
- If nothing provided, shows gentle message: "Please upload a file, type a topic, or paste text"

Screen 2: Reading

What Happens Behind The Scenes

For File Upload:

Plaintext
```
Step 1: Detect file type
Step 2: Extract text from file
  - PDF → text extraction page by page
  - Images inside PDF → OCR to read text from images
  - DOC/DOCX → parse document structure
  - TXT/MD → read directly
  - HTML → strip tags, keep content
  - JPG/PNG → full OCR
Step 3: Clean up junk
  - Remove watermarks ("studyiq.com", "visit:", etc.)
  - Remove page numbers ("Page no. 1"
  - Remove repeated headers/footers
  - Remove text shorter than 10 characters
Step 4: Organize in reading order
  - Sort content top to bottom per page
  - Merge text and image-extracted text
  - Maintain page references
```

For Topic Name:

Plaintext
```
tep 1: AI generates comprehensive content on the topic
Step 2: Content includes definitions, examples, comparisons, data
Step 3: Structured with clear headings
Step 4: Treated as if it were an uploaded document
```

For Pasted Text:

Plaintext
```
Step 1: Take text as-is
Step 2: Try to detect structure (headings, lists)
Step 3: Treat as single-page document
```

What User Sees

Plaintext
```
┌──────────────────────────────────────┐
│ 📖 Reading your material...          │
├──────────────────────────────────────┤
│                                      │
│ ✅ File accepted (National_Income.pdf)│
│ ✅ Extracted 22 pages of text        │
│ ✅ Found 6 images — reading text from them │
```

```
│ ✅ Cleaned up noise and watermarks      │
│ ⏳ Understanding the content...          │
│                                         │
│ [████████████░░░░░░░░ 65%]              │
│                                         │
│ This may take 30-60 seconds             │
└─────────────────────────────────────────┘
```

- Each step shows checkmark when complete
- Progress bar moves smoothly
- No user action needed
- Auto-advances to Screen 3 when done

Screen 3: Understanding + Feature Detection

What AI Analyzes
The AI reads ALL extracted text and produces a comprehensive understanding.

For PSC Content (National Income Example)

What AI Detects:

- Subject: Economics
- Sub-discipline: Macroeconomics — National Income Accounting
- Purpose: Teach measurement of national economic output
- Topics: National Income, GDP, GNP, NDP, NNP, NI, PI, DPI, Per Capita Income, GDP Deflator, Base Year, GDP Calculation India, GDP-GVA Gap, Potential GDP, GDP & Welfare, Green GDP, GNHI, HDI, Thriving Places Index (Count: 19)
- Definitions: National Income, GDP, GNP, NDP, NNP, Factor Cost, Market Price, Depreciation, Transfer Payments, Domestic Territory, Residents, Value Addition, etc. (Count: 20+)
- Formulas: GDP = C+I+G+(X-M), GNP = GDP+NFIA, NDP = GDP-Depreciation, NNP = GNP-Depreciation, MP = FC+Taxes-Subsidies, GDP Deflator = Nominal/Real×100, Per Capita = NI/Population (Count: 12+)
- Tables: Factor Cost vs Market Price, GDP vs GNP, Old NAS vs New NAS, GVA vs GDP, Comparison Matrix, HDI Significance (Count: 6)
- Examples: Mr. Zen resident example, Shyam bread value addition, Shyam orange juice, Maggi noodles, LPG cylinder pricing, Mr. Y stocks nominal/real, Car production GDP (Count: 7)
- PYQs: Multi-dimensional Poverty Index (2012), National Income definition (2013), GNP economic development (2018), Potential GDP Mains (2020) (Count: 4)
- Diagrams: Basic concepts circle diagram, Nominal vs Real Value infographic, Per Capita Income bar chart, GDP Growth chart, GNH Domains wheel, HDI Components diagram (Count: 6)
- Comparisons: GDP vs GNP, Nominal vs Real GDP, Factor Cost vs Market Price, GVA vs GDP, Old NAS vs New NAS (Count: 5)
- Timelines: Per Capita Income India 1950-2021, Base Year changes, NAS evolution (Count: 3)

Content Layers Detected:

- CONCEPTUAL: High Confidence (20+ definitions with explanations)
- FORMULA_MATHEMATICAL: High Confidence (12+ formulas with variables)
- COMPARISON: High Confidence (5 explicit comparison tables)
- CHRONOLOGICAL: Medium Confidence (Per capita income timeline, base year history)
- STATISTICAL_DATA: High Confidence (GDP growth rates, PCI data)
- PYQ_VERIFIED: High Confidence (4 questions with years and answers)
- NARRATIVE_SCENARIO: Medium Confidence (Character-based examples - Shyam, Mr. Zen)

For Computer Science Content (Example: Data Structures)

What AI Detects:

- Subject: Computer Science
- Sub-discipline: Data Structures and Algorithms
- Purpose: Teach fundamental data structures with implementation
- Topics: Arrays, Linked Lists, Stacks, Queues, Trees, Graphs, Hash Tables, Sorting, Searching (Count: 9)
- Definitions: Array, Node, Pointer, Stack, Queue, Tree, Graph, Hash Function, Big-O, etc. (Count: 15+)
- Code Blocks: Implementation examples in Python/Java/C++ (Count: 20+)
- Complexity Tables: Time/Space for each operation on each structure (Count: 5+)
- Diagrams: Tree structure, Graph representations, Stack operations, Queue operations (Count: 10+)
- Input/Output Demos: "If input is [1,3,2] → sorted output is [1,2,3]" (Count: 8+)
- Comparisons: Array vs Linked List, BFS vs DFS, Stack vs Queue (Count: 4+)

Content Layers Detected:

- CONCEPTUAL: High Confidence (Definitions and explanations)
- CODE_IMPLEMENTATION: High Confidence (Multiple code blocks)
- ALGORITHM_ANALYSIS: High Confidence (Complexity tables)
- COMPARISON: High Confidence (Structure vs structure tables)
- VISUAL_STRUCTURAL: High Confidence (Tree/Graph diagrams)
- INPUT_OUTPUT_DEMO: Medium Confidence (Example runs with data)

Feature Detection Display

After analysis, the app shows detected features as toggleable items.
For PSC Content:

Plaintext

```
┌─────────────────────────────────────────┐
│ 📚 Subject: Economics (Macroeconomics)   │
│ 📋 Purpose: National Income Accounting   │
│                                         │
│ Found in your material:                 │
│ • 19 topics  • 20 definitions  • 12 formulas │
│ • 4 PYQs   • 6 diagrams   • 5 comparisons │
│ • 7 examples • 3 timelines   • 6 tables  │
│                                         │
│ [👁️ See full details]                    │
│                                         │
│ ── Features Detected (toggle on/off) ── │
│                                         │
│ ✅ Current Affairs     (economics topic) │
│ ✅ State-Specific Data   (MP focus set)  │
│ ✅ PYQ Integration     (4 PYQs found)    │
│ ✅ Formula Breakdown    (12 formulas)    │
│ ✅ Comparison Tables    (5 comparisons)  │
│ ✅ Timeline/History     (3 timelines)    │
│ ✅ Story/Scenario Boxes   (7 examples)   │
│ ✅ Legal/Constitutional  (policy related) │
│ ❌ Code Examples        (not applicable) │
│ ❌ Algorithm Analysis    (not applicable) │
│                                         │
│ [Show more options ▼]                   │
│                                         │
│ [🔄 Re-analyze]  [✏️ Fix manually]        │
│ [Looks good → Continue]                 │
└─────────────────────────────────────────┘
```

For Computer Science Content:

Plaintext

```
┌─────────────────────────────────────────┐
│ 📚 Subject: Computer Science             │
│ 📋 Purpose: Data Structures & Algorithms │
│                                         │
│ Found in your material:                 │
│ • 9 topics   • 15 definitions  • 20 code blocks│
│ • 5 complexity tables  • 10 diagrams     │
│ • 8 input/output demos  • 4 comparisons  │
│                                         │
│ ── Features Detected (toggle on/off) ── │
│                                         │
│ ✅ Code Blocks         (20 found)        │
│ ✅ Algorithm Complexity   (5 tables)     │
│ ✅ Input → Output Demos   (8 examples)   │
│ ✅ Data Structure Visuals (10 diagrams)  │
│ ✅ Comparison Tables     (4 comparisons) │
│ ✅ Architecture Diagrams  (structure visuals) │
│ ❌ Current Affairs      (not needed)     │
│ ❌ State Focus         (not applicable)  │
│ ❌ PYQ Integration      (none found)     │
```

```
│ ❌ Legal/Constitutional   (not applicable)    │
│                                               │
│ [Show more options ▼]                         │
│                                               │
│ [Looks good → Continue]                       │
└───────────────────────────────────────────────┘
```

"See Full Details" — Expandable Section

When user clicks "See full details", shows:
Definitions found:

ID	Definition	Page
D-001	National Income means the total money value of all final goods and services...	3
D-002	GDP is the total market value of final goods and services produced within the country...	4
D-003	GNP is the total value of final goods and services produced by citizens of a country...	10

Formulas found:

ID	Formula	Variables	Page
F-001	GDP at MP = C + I + G + NX(X-M)	C=Consumption, I=Investment, G=Govt, X=Export, M=Import	10
F-002	GNP = GDP + NFIA	NFIA = Net Factor Income from Abroad	11

PYQs found:

ID	Year	Type	Question (first 50 chars)	Answer
PYQ-001	2012	Prelims	The Multi-dimensional Poverty Index developed by...	(a)
PYQ-002	2013	Prelims	The national income of a country for a given per...	(a)
PYQ-003	2018	Prelims	Increase in absolute and per capita real GNP do...	(c)
PYQ-004	2020	Mains	Define potential GDP and its determinants...	10 Marks

STAGE 2: PLAN THE BLUEPRINT

Purpose

For each part, create a detailed teaching plan showing exactly what will be written, which images will be created, where current affairs will go, and how validation will work.

Screen 4: Parts

What AI Does
Takes the analysis from Stage 1 and divides the content into logical teachable parts.
Part Count Rules

Total Pages	Parts Range
1-10	1-3 parts
11-20	2-6 parts

21-30	4-8 parts
31-50	5-10 parts
51-100	8-15 parts
101-200	12-20 parts

Splitting Rules

- Conceptual coherence first — related topics stay together
- Never split a formula from its explanation
- PYQs go with the topic they test
- Never create a tiny part with just 1-2 paragraphs
- Never merge completely unrelated topics
- Each part must make sense on its own

For PSC Content (National Income — 22 pages)

AI suggests 6 parts:

Plaintext

```
┌──────────────────────────────────────────────┐
│ 📄 How I'll split your material (6 parts)     │
├──────────────────────────────────────────────┤
│                 │
│ ┌─ Part 1 ──────────────────────────────┐ │
│ │ 📝 Foundations of National Income       │ │
│ │ Pages 3-5 | 3 topics | 6 definitions    │ │
│ │ What NI means, why it matters, problems  │ │
│ └───────────────────────────────────────┘ │
│                 │
│ ┌─ Part 2 ──────────────────────────────┐ │
│ │ 📝 Types of GDP & Measurement Methods    │ │
│ │ Pages 6-10 | 4 topics | 5 formulas      │ │
│ │ Nominal/Real GDP, Value Added, Income,   │ │
│ │ Expenditure methods                      │ │
│ └───────────────────────────────────────┘ │
│                 │
│ ┌─ Part 3 ──────────────────────────────┐ │
│ │ 📝 NDP, GNP, NNP & Other Measures        │ │
│ │ Pages 10-13 | 5 topics | 4 formulas     │ │
│ │ All income measures, Personal Income,    │ │
│ │ Per Capita Income                        │ │
│ └───────────────────────────────────────┘ │
│                 │
│ ┌─ Part 4 ──────────────────────────────┐ │
│ │ 📝 GDP Deflator, Base Year, India's GDP  │ │
│ │ Pages 13-16 | 4 topics | 1 formula      │ │
│ │ How India calculates GDP, NAS changes,   │ │
│ │ GVA vs GDP gap                           │ │
│ └───────────────────────────────────────┘ │
│                 │
│ ┌─ Part 5 ──────────────────────────────┐ │
│ │ 📝 GDP, Welfare & Alternative Measures   │ │
│ │ Pages 17-20 | 4 topics | 0 formulas     │ │
│ │ Green GDP, GNHI, limitations of GDP      │ │
│ └───────────────────────────────────────┘ │
│                 │
│ ┌─ Part 6 ──────────────────────────────┐ │
│ │ 📝 HDI, Indices & Previous Year Questions │ │
│ │ Pages 20-22 | 3 topics | 4 PYQs         │ │
│ │ HDI components, Thriving Places, PYQs    │ │
│ └───────────────────────────────────────┘ │
│                 │
│ [➕ Add Part] [🔀 Merge Parts] [✏️ Edit]     │
│ [Continue to Blueprint →]                   │
└──────────────────────────────────────────────┘
```

For Computer Science Content (Data Structures)

AI suggests 4 parts:

Plaintext

```
┌─ Part 1 ──────────────────────────┐
│ 📝 Linear Structures              │
│ Arrays, Linked Lists              │
│ 8 code blocks, 2 complexity tables │
└────────────────────────────────────┘

┌─ Part 2 ──────────────────────────┐
│ 📝 Stack & Queue                  │
│ Implementation, applications      │
│ 6 code blocks, 4 I/O demos        │
└────────────────────────────────────┘

┌─ Part 3 ──────────────────────────┐
│ 📝 Trees & Graphs                 │
│ BST, BFS, DFS, traversals         │
│ 10 code blocks, 6 diagrams        │
└────────────────────────────────────┘

┌─ Part 4 ──────────────────────────┐
│ 📝 Searching & Sorting            │
│ All algorithms, comparisons       │
│ 8 code blocks, master comparison  │
└────────────────────────────────────┘
```

Screen 5: Blueprint (Card Grid)

What User Sees

Grid of blueprint cards — one per part. Each card shows a compact summary.
PSC Blueprint Card Example (Part 2)

Plaintext
```
┌─────────────────────────────────────────────┐
│ 📝 Part 2: Types of GDP & Measurement Methods │
├─────────────────────────────────────────────┤
│                │
│ 📖 Text Plan:                │
│ • Nominal vs Real GDP with examples        │
│ • Value Addition method (Shyam bread story)   │
│ • Income method (4 factors of production)    │
│ • Expenditure method (C+I+G+NX formula)      │
│                │
│ 🎨 Images Planned: 5               │
│ • Nominal vs Real comparison infographic     │
│ • Shyam bread value addition comic        │
│ • 3 methods flowchart              │
│ • GDP formula breakdown pictograph        │
│ • Factor income pie chart            │
│                │
│ 📰 Current Affairs: 2 search queries        │
│ 🔍 Validation: Planned              │
│                │
│ [Click to see full blueprint →]           │
│ [✅ Approve this Part]              │
│                │
└─────────────────────────────────────────────┘
```

Master Controls
[✅ Approve All Blueprints] ← approves every part at once
Each card also has individual approve button.

Screen 6: Blueprint Detail View

When user clicks a card, the full-screen detail view opens.
Layout

Plaintext
```
┌──────────────────────────────────────────────────────────────┐
│ [← Back to Blueprint Cards]                    │
│                                 │
│ Part 2: Types of GDP & Measurement Methods              │
├──────────────────────────────────┬───────────────────────────┤
│                 │                     │
│ LEFT SIDE (70%)           │ RIGHT SIDE (30%)      │
│                 │ STICKY AI CHAT        │
│ ┌── SECTION 1: HEADINGS ──────────┐ │                  │
│ │               │ │ ┌──────────────────┐     │
│ │ Main: Types of GDP &      │ │ │ 💬 AI Assistant │   │
│ │    Measurement Methods   │ │ │          │   │
│ │               │ │ │ Ask me to modify │   │
│ │ Sub-headings:         │ │ │ anything about  │   │
│ │ 1. Nominal GDP vs Real GDP   │ │ │ this part's plan │   │
│ │ 2. Real and Nominal Value   │ │ │          │   │
│ │ 3. Production/Value Added    │ │ │ [Type here...]   │   │
│ │ 4. Income/Factor Earning    │ │ │ [Send]       │   │
│ │ 5. Expenditure Method      │ │ │          │   │
│ │               │ │ │ Examples:      │   │
│ └─────────────────────────────────┘ │ │ "Add more focus   │   │
│                 │ │  on formulas"    │   │
│ ┌── SECTION 2: TEXT PLAN ─────────┐ │ │ "Remove timeline"│   │
│ │               │ │ │ "Add MP specific │   │
│ │ What will be written:      │ │ │  GSDP comparison"│   │
│ │               │ │ │ "Change image    │   │
│ │ 📗 From your PDF:        │ │ │  style to comic"│   │
│ │ • Nominal GDP definition and  │ │ │          │   │
│ │  calculation at current price │ │ │ [Minimize ▼]   │   │
│ │ • Real GDP at constant price  │ │ └──────────────────┘     │
│ │ • Car production example    │ │              │
│ │ • Shyam bread value addition  │ │              │
│ │ • All 3 methods with formulas  │ │              │
│ │               │ │              │
│ │ 🔍 Will be researched/added:  │ │              │
│ │ • Latest GDP data for India   │ │              │
│ │ • MP GSDP if available     │ │              │
│ │ • Simple prerequisite on    │ │              │
│ │  "What is an economy"     │ │              │
│ │  (if basic-to-advanced mode)  │ │              │
│ │               │ │              │
│ │ 💎 Optional enrichment:     │ │              │
│ │ • Global GDP ranking table   │ │              │
│ │ • Fun fact about GDP history  │ │              │
│ │               │ │              │
│ │ Containers planned:       │ │              │
│ │ ☑ Formula Derivation Box    │ │              │
│ │ ☑ Story/Scenario Box      │ │              │
│ │ ☑ Comparison Table       │ │              │
│ │ ☑ Common Confusion Alert    │ │              │
│ │ ☐ Legal/Case Law Box      │ │              │
│ │ ☐ Timeline Strip        │ │              │
│ │               │ │              │
│ │ Generation Prompt (editable): │ │              │
│ │ ┌───────────────────────────┐ │ │              │
│ │ │ Write comprehensive    │ │ │              │
│ │ │ notes on Types of GDP   │ │ │              │
│ │ │ covering Nominal vs    │ │ │              │
│ │ │ Real, all 3 methods...   │ │ │              │
│ │ │ [editable text area]    │ │ │              │
│ │ └───────────────────────────┘ │ │              │
│ │ [📋 Copy] [🔄 Regenerate]    │ │              │
│ │               │ │              │
│ └─────────────────────────────────┘ │              │
│                 │              │
│ ┌── SECTION 3: IMAGE PLAN ────────┐ │              │
│ │               │ │              │
│ │ Image 1 of 5:         │ │              │
│ │ 🎨 Nominal vs Real Comparison │ │              │
│ │ Style: Infographic (auto-pick) │ │              │
│ │ [Change style ▼]       │ │              │
│ │  Flowchart          │ │              │
│ │  Comic Story         │ │              │
│ │  Comparison Matrix      │ │              │
│ │  Venn Diagram         │ │              │
│ │  ... (all 35 styles)     │ │              │
│ │               │ │              │
│ │ What this image will show:   │ │              │
│ │ "Left side: Nominal value —   │ │              │
│ │  what you see in supermarket, │ │              │
│ │  bank account, pay slips.   │ │              │
│ │  Right side: Real value —   │ │              │
│ │  adjusted for inflation,    │ │              │
│ │  better for comparisons"    │ │              │
│ │               │ │              │
```

```
│ │ Labels: Nominal Value, Real   │ │
│ │ Value, Prices, Inflation,     │ │
│ │ Purchasing Power              │ │
│ │                               │ │
│ │ Prompt preview:               │ │
│ │ ┌───────────────────────────┐ │ │
│ │ │ Create a split-screen     │ │ │
│ │ │ infographic comparing     │ │ │
│ │ │ nominal vs real value     │ │ │
│ │ │ [editable]                │ │ │
│ │ └───────────────────────────┘ │ │
│ │                               │ │
│ │ ── Image 2 of 5 ──            │ │
│ │ 🎨 Shyam Bread Value Addition │ │
│ │ Style: Comic Story (auto-pick)│ │
│ │                               │ │
│ │ What this image will show:    │ │
│ │ "4-panel comic:               │ │
│ │  Panel 1: Miller sells flour  │ │
│ │  to Shyam for ₹50             │ │
│ │  Panel 2: Shyam bakes bread   │ │
│ │  Panel 3: Shyam sells bread   │ │
│ │  for ₹100                     │ │
│ │  Panel 4: Value Added = ₹50   │ │
│ │  (₹100 - ₹50)"                │ │
│ │                               │ │
│ │ [Change style ▼] [Edit prompt]│ │
│ │                               │ │
│ │ ── Image 3, 4, 5... ──        │ │
│ │                               │ │
│ └───────────────────────────────┘ │
│                                   │
│ ┌── SECTION 4: CA PLAN ─────────┐ │
│ │                               │ │
│ │ Current Affairs Search Plan:  │ │
│ │                               │ │
│ │ Search 1: "India GDP growth   │ │
│ │ rate 2024 latest quarterly"   │ │
│ │ Purpose: Latest GDP number    │ │
│ │ [Edit query]                  │ │
│ │                               │ │
│ │ Search 2: "India GDP method   │ │
│ │ change 2020 2021 2022 2023"   │ │
│ │ Purpose: Major methodological │ │
│ │ changes in GDP calculation    │ │
│ │ [Edit query]                  │ │
│ │                               │ │
│ │ Search 3: "Madhya Pradesh GSDP│ │
│ │ 2024 growth economic"         │ │
│ │ Purpose: MP state data        │ │
│ │ [Edit query]                  │ │
│ │                               │ │
│ │ Search 4: "MP economic        │ │
│ │ achievement 2020 2021 2022"   │ │
│ │ Purpose: MP major news        │ │
│ │ [Edit query]                  │ │
│ │                               │ │
│ └───────────────────────────────┘ │
│                                   │
│ ┌── SECTION 5: VALIDATION PLAN ──┐│
│ │                               │ │
│ │ After generation, AI will:    │ │
│ │ ☑ Check all formulas correct  │ │
│ │ ☑ Verify facts against source │ │
│ │ ☑ Review images match content │ │
│ │ ☑ Suggest missing visuals     │ │
│ │ ☑ Check CA is properly placed │ │
│ │                               │ │
│ └───────────────────────────────┘ │
│                                   │
│ [✅ Approve this Part's Blueprint] │
│ [💾 Save Changes]                  │
│                                   │
└───────────────────────────────────┴──────────────────────┘
```

AI Chat Examples (Right Panel)
User can type things like:

User Says	AI Does
"Add more focus on formulas"	Adds more formula breakdown boxes to text plan
"Remove the comic for Shyam, make it a flowchart"	Changes Image 2 style from Comic to Flowchart
"Add MP GSDP comparison with India GDP"	Adds new image plan + CA search query
"I want a timeline of base year changes"	Adds timeline image to image plan
"Remove current affairs for this part"	Removes CA plan section
"Add a misconception alert about GDP vs GNP"	Adds Common Confusion container to text plan
"Make this part simpler, I'm a beginner"	Adjusts text plan to add prerequisite explanations

Computer Science Blueprint Detail View — Differences
For a CS part (e.g., "Stack & Queue"), the blueprint would show:
Text Plan would include:

Plaintext

```
📗 From your material:
• Stack definition and LIFO principle
• Push, Pop, Peek operations
• Queue definition and FIFO principle
• Enqueue, Dequeue operations

🔍 Will be added:
• Real-world analogies (stack of plates, queue at ticket counter)
• Common interview questions on stacks/queues
• Edge cases (overflow, underflow)

Containers planned:
☑ Code Block (implementation)
☑ Input → Output Demo
☑ Complexity Table
☑ Comparison Table (Stack vs Queue)
☑ Visual Diagram (operations)
☐ Current Affairs (not applicable)
☐ Legal Context (not applicable)
```

Image Plan would include:

Plaintext

```
Image 1: Stack Operations Visualization
Style: Step-by-Step Sequence
Shows: Push(5) → Push(3) → Pop() → result
Labels: Stack, Top, Push, Pop, LIFO

Image 2: Queue Operations Visualization
Style: Step-by-Step Sequence
Shows: Enqueue(A) → Enqueue(B) → Dequeue() → result
Labels: Queue, Front, Rear, FIFO

Image 3: Stack vs Queue Comparison
Style: Comparison Matrix + Mixed Diagram
Shows: Side-by-side comparison with visual + table

Image 4: Real-World Applications
Style: Infographic
Shows: Browser back button (stack), Printer queue (queue),
    Function call stack, BFS queue
```

STAGE 3: WRITE & GENERATE

Purpose

Execute the approved blueprints. Actually write the notes, create images, search current affairs, validate quality. Everything happens on one long scrollable page.

Screen 7: The Generation Workspace

Overall Layout

Plaintext

```
┌──────────────────────────────────────────────────────────────┐
│ TOP BAR                                      │
│ [Generate All] [Progress: 1/6 parts done] [████░░░░░░ 17%]     │
├──────────┬───────────────────────────────────────────────────┤
│        │                              │
│ SIDEBAR │ MAIN CONTENT AREA                        │
│        │ (scrollable, all parts stacked)              │
│ Part 1 ✅│                                │
│ Part 2 🔄│ ══════════════════════════════════════════════   │
│ Part 3 ⏳│ PART 1: Foundations of National Income          │
│ Part 4 ⏳│ ══════════════════════════════════════════════   │
│ Part 5 ⏳│                                │
│ Part 6 ⏳│ [Generate Part 1] [Approve Part 1]             │
│        │                              │
│ ────────── │ ... content ...                          │
│ Progress │                              │
│ 1/6 done │ ══════════════════════════════════════════════   │
│        │ PART 2: Types of GDP & Methods                │
│        │ ═════════════════════════════════════════════   │
│        │                              │
│        │ [Generate Part 2] [Approve Part 2]             │
│        │                              │
│        │ ... and so on for all parts ...               │
│        │                              │
└──────────┴───────────────────────────────────────────────────┘
```

Sidebar Behavior

- ⏳ Pending — not generated yet
- 🔄 Currently generating
- 📝 Generated — awaiting review
- ✅ Approved by user
- 🔓 Unlocked for editing (was approved)

Click any part name → scrolls to that part section.
If more than 10 parts, sections start collapsed. Click to expand.

What Happens When User Clicks "Generate Part 1"

Plaintext

```
Step 1: Text Generation
  AI writes comprehensive notes using the blueprint plan
  Text streams in real-time (user can watch it appear)
  Containers render as styled boxes
  IMAGE_NEEDED tags appear as placeholder cards

Step 2: Image Generation
  For each planned image, AI builds the prompt
  Image generates using Gemini's built-in image generation
  Image appears inline where the placeholder was

Step 3: Current Affairs (if applicable)
  CA search buttons appear at relevant positions
  Not auto-triggered — user clicks when ready

Step 4: Ready for review
  Part section fully populated
  All interactive controls active
```

Single Part — Full Detail View

Here is what ONE generated part looks like on the page:

Plaintext

```
════════════════════════════════════════════════════
PART 2: Types of GDP & Measurement Methods
[Generate] [Approve Text ☐] [Approve Images ☐] [Approve Part]
════════════════════════════════════════════════════

┌─ 💡 Concept Clarification ─────────────────────┐
│                                                 │
│ Before understanding types of GDP, remember:    │
│ GDP measures the total market value of final goods │
│ and services produced within the country's borders │
│ during a year.                                  │
│                          [SOURCE_BACKED] │
└─────────────────────────────────────────────────┘

## Nominal GDP (GDP at Current Prices)

The total monetary value of all goods and services
produced in the domestic territory, counted on the
basis of market prices prevailing in that year.

Key points:                    [SOURCE_BACKED]
• Calculated at current market price
• Does NOT adjust for inflation
• Can be misleading for growth comparison

┌─ 📊 Formula Box ────────────────────────────────┐
│                                                 │
│ Nominal GDP = Σ (Current Year Price × Current Year │
│            Quantity)                            │
│                                                 │
│ Example:                                        │
│ If a country produces 300 cars at ₹110 each in 2022-23 │
│ Nominal GDP = 300 × 110 = ₹33,000     [SOURCE_BACKED]│
└─────────────────────────────────────────────────┘

## Real GDP (GDP at Constant Prices)

The total monetary value of all final goods and services
valued at a constant base year price.
                         [SOURCE_BACKED]
Key points:
• Calculated at base year prices
• IS adjusted for inflation
• Better measure for comparing economic growth over time

┌─ 📊 Formula Box ────────────────────────────────┐
│                                                 │
│ Real GDP = Σ (Base Year Price × Current Year Quantity) │
│                                                 │
│ Using same example:                             │
│ Base year 2011-12 price = ₹100 per car          │
│ Real GDP = 300 × 100 = ₹30,000       [SOURCE_BACKED]│
└─────────────────────────────────────────────────┘

┌─ ⚠️ Common Confusion Alert ─────────────────────┐
│                                                 │
│ Students often confuse:                         │
│ • Nominal GDP growing does NOT mean economy is growing │
│ • Prices might just be rising (inflation)       │
│ • ALWAYS check Real GDP for actual growth       │
│                    [OPTIONAL_ENRICHMENT] │
└─────────────────────────────────────────────────┘

🖼️ ┌───────────────────────────────────────────────┐
 │                                                 │
 │ [GENERATED IMAGE: Nominal vs Real Value Infographic]│
 │ Shows: Supermarket prices, bank account (Nominal) │
 │     vs Inflation-adjusted purchasing power (Real)│
 │                                                 │
 │ [🔄 Regenerate] [🔄 Different Style ▼] [✏️ Edit Prompt]│
 │ [✅ Approve Image]                              │
 └─────────────────────────────────────────────────┘

## Value Addition Method (Production Method)

National income is estimated by calculating the value
addition done by firms at each stage of production.
                         [SOURCE_BACKED]

**Value Added = Value of Output − Intermediate Consumption**

┌─ 📖 Story Box ──────────────────────────────────┐
│                                                 │
│ 🧑 Meet Shyam — The Bread Maker                 │
│                                                 │
│ Step 1: Shyam buys flour from a miller for ₹50  │
│      (This is intermediate consumption)         │
│                                                 │
│ Step 2: Shyam bakes the flour into bread        │
│      (Production activity)                      │
│                                                 │
│ Step 3: Shyam sells the bread for ₹100          │
│      (This is the value of output)              │
│                                                 │
│ Step 4: Value Added by Shyam = ₹100 - ₹50 = ₹50 │
│                                                 │
│ The ₹50 is what Shyam truly created. The other ₹50 │
│ was the miller's contribution.                  │
│                          [SOURCE_BACKED] │
└─────────────────────────────────────────────────┘

🖼️ ┌───────────────────────────────────────────────┐
 │                                                 │
 │ [GENERATED IMAGE: Shyam Bread Comic - 4 panels] │
 │ Panel 1: Miller gives flour bag "₹50"           │
 │ Panel 2: Shyam baking with oven                 │
 │ Panel 3: Customer buying bread "₹100"           │
 │ Panel 4: Caption "Value Added = ₹50!"           │
 │                                                 │
 │ [🔄 Regenerate] [🔄 Different Style ▼] [✏️ Edit Prompt]│
 │ [✅ Approve Image]                              │
 └─────────────────────────────────────────────────┘

## Expenditure Method

GDP is estimated by the expenditure made on the
purchase of final goods and services.
                         [SOURCE_BACKED]

┌─ 📊 Formula Box ────────────────────────────────┐
│                                                 │
```

```
│ GDP at MP = C + I + G + NX(X-M)                     │
│                                                      │
│ Where:                                               │
│ C  = Consumption Expenditure (households buying      │
│      TV, car, food, etc.)                            │
│ I  = Investment Expenditure (machinery, buildings,   │
│      residential property, highways)                 │
│ G  = Government Expenditure (admin, defence,         │
│      law and order — NOT scholarships/subsidies)     │
│ NX = Net Exports = Exports(X) - Imports(M)           │
│                                                      │
│ ⚠️ NOT counted in I: savings in bank, shares, bonds  │
│ ⚠️ NOT counted in G: scholarships, subsidies,        │
│    transfer payments                                 │
│                              [SOURCE_BACKED]         │
└──────────────────────────────────────────────────────┘

─── Current Affairs Zone ─────────────────────────────────

📰 [🔍 Generate Current Affairs for "GDP Methods"]

  When clicked, searches and shows:

  ┌─ 📰 National Recent (2024-Today) ──────────────────┐
  │                                                     │
  │ • India GDP grew 6.7% in Q2 FY25 (MoSPI, Nov 2024)  │
  │ • GDP growth at 8.2% for FY2023-24 (Final estimate) │
  │                                                     │
  │ [➕ Merge into notes above] [Merge at specific spot ▼]│
  └─────────────────────────────────────────────────────┘

  ┌─ 📰 National Major (2020-2023) ─────────────────────┐
  │                                                     │
  │ • India became 5th largest economy (2022)           │
  │ • GDP contracted -6.6% during COVID (2020-21)       │
  │ • Base year revision proposal 2017-18 (2023)        │
  │                                                     │
  │ [➕ Merge into notes]                                │
  └─────────────────────────────────────────────────────┘

  ┌─ 📰 MP State (2024-Today) ──────────────────────────┐
  │                                                     │
  │ • MP GSDP growth at 7.1% (2024 estimate)            │
  │ • [CA_STATUS: Limited state-specific data found]    │
  │                                                     │
  │ [➕ Merge into notes]                                │
  └─────────────────────────────────────────────────────┘

  ┌─ 📰 MP State Major (2020-2023) ─────────────────────┐
  │                                                     │
  │ • MP ranked among top 5 states in GDP growth (2022) │
  │                                                     │
  │ [➕ Merge into notes]                                │
  └─────────────────────────────────────────────────────┘

─── User Highlighting → Image Generation ─────────────────

When user selects text "GDP at MP = C + I + G + NX(X-M)"
and its explanation:

🎨 [Visualize This] button appears

→ User clicks → popup shows:

┌─ 🎨 Visualize Selected Text ───────────────────────────┐
│                                                        │
│ Selected: "GDP at MP = C + I + G + NX(X-M)..."         │
│                                                        │
│ Auto-generated prompt:                                 │
│ ┌──────────────────────────────────────────────────┐   │
│ │ Create an infographic showing GDP formula        │   │
│ │ breakdown. Center: GDP bubble. Four branches:    │   │
│ │ C (Consumption) with household icons,            │   │
│ │ I (Investment) with factory icon,                │   │
│ │ G (Government) with parliament icon,             │   │
│ │ NX (Net Export) with ship icon.                  │   │
│ │ Below each: what's included, what's NOT.         │   │
│ │ [EDITABLE - user can modify this prompt]         │   │
│ └──────────────────────────────────────────────────┘   │
│                                                        │
│ Recommended styles:                                    │
│ ★ Pictograph/Icon Array — best for formula components  │
│ ○ Mind Map — shows relationships                       │
│ ○ Infographic — detailed visual                        │
│                                                        │
│ [Change to any style ▼] (all 35 styles dropdown)       │
│                                                        │
│ OR describe your own vision:                           │
│ ┌──────────────────────────────────────────────────┐   │
│ │ I want a pie chart showing rough percentage      │   │
│ │ of each component for India...                   │   │
│ │ [AI will auto-build prompt from your description]│   │
│ └──────────────────────────────────────────────────┘   │
│                                                        │
│ [Generate Image] [Cancel]                              │
└────────────────────────────────────────────────────────┘

After generation:

┌─ 🖼️ Generated Image Preview ───────────────────────────┐
│                                                        │
│ [The actual generated image appears here]              │
│                                                        │
│ [✅ Insert Here] [🔄 Regenerate Same Style]            │
│ [🔄 Try Different Style ▼] [✏️ Edit Prompt & Retry]    │
│ [❌ Discard]                                            │
└────────────────────────────────────────────────────────┘

─── AI Validation ────────────────────────────────────────

[🔍 AI Validate This Part]

When clicked, AI reviews everything and shows:

┌─ 🔍 AI Validation Results ─────────────────────────────┐
│                                                        │
│ Overall Score: ✅ Good                                 │
│                                                        │
│ ✅ All 5 formulas correctly presented                  │
│ ✅ All examples match source material                  │
│ ✅ Story box is engaging and accurate                  │
│ ⚠️ Image 3 (3 methods flowchart) could include         │
│    a small data table below each method                │
```

```
│     [Accept & Regenerate Image] [Ignore]              │
│  ⚠️ Missing: Comparison table for 3 methods            │
│     (which method used where?)                        │
│     [Accept & Generate Table] [Ignore]                │
│  💡 Suggestion: Add a "Quick Revision" box at end      │
│     summarizing all formulas                          │
│     [Accept & Add] [Ignore]                           │
│                                                       │
└───────────────────────────────────────────────────────┘

──── Refinement Options ──────────────────────────────────

[🔄 Regenerate Entire Text]
[✏️ Inline Edit] ← makes text directly editable
[💬 AI Refine] ← opens chat:

┌─ 💬 Refinement Chat ───────────────────────────────────┐
│                                                       │
│  You: "Make the formula explanation simpler,          │
│        a 10th grader should understand"               │
│                                                       │
│  AI: "I'll rewrite the formula section with           │
│       simpler language and a household budget         │
│       analogy..."                                     │
│  [text updates in the notes above]                    │
│                                                       │
│  You: "Add MP comparison wherever India data exists"  │
│                                                       │
│  AI: "Adding MP GSDP data alongside India GDP data    │
│       in all relevant sections..."                    │
│  [text updates]                                       │
│                                                       │
│  [Type here...] [Send]                                │
└───────────────────────────────────────────────────────┘

──── Approval Controls ───────────────────────────────────

[Approve Text ☐] [Approve Images ☐]
[✅ Approve Entire Part 2]

After approving:

┌─── Part 2 ── ✅ APPROVED ───────────────────────────────┐
│                                                       │
│  [All content visible but locked/grayed slightly]     │
│                                                       │
│  [🔓 Unlock to Edit]                                   │
│                                                       │
└───────────────────────────────────────────────────────┘

═══════════════════════════════════════════════════════════
SEPARATOR — NEXT PART BEGINS
═══════════════════════════════════════════════════════════
PART 3: NDP, GNP, NNP & Other Measures
[Generate Part 3] [Approve Part 3]
═══════════════════════════════════════════════════════════

... same structure repeats for each part ...
```

Computer Science — Stage 3 Differences

For a CS part (Stack & Queue), the generated content would look different:

````
═══════════════════════════════════════════════════════════
PART 2: Stack & Queue
═══════════════════════════════════════════════════════════

## Stack

A Stack is a linear data structure that follows the
Last In First Out (LIFO) principle.
                              [SOURCE_BACKED]

Think of it like a stack of plates — you can only
add or remove from the top.        [OPTIONAL_ENRICHMENT]

┌─ 💻 Code Block ─────────────────────────────────────────┐
│                                                       │
│  ```python                                            │
│  class Stack:                                         │
│      def __init__(self):                              │
│          self.items = []                              │
│                                                       │
│      def push(self, item):                            │
│          self.items.append(item)                      │
│                                                       │
│      def pop(self):                                   │
│          if not self.is_empty():                      │
│              return self.items.pop()                  │
│                                                       │
│      def peek(self):                                  │
│          return self.items[-1]                        │
│                                                       │
│      def is_empty(self):                              │
│          return len(self.items) == 0                  │
│  ```                                                  │
│                              [SOURCE_BACKED]          │
└───────────────────────────────────────────────────────┘

┌─ 🔄 Input → Output Demo ───────────────────────────────┐
│                                                       │
│  >>> s = Stack()                                      │
│  >>> s.push(5)                                        │
│  >>> s.push(3)                                        │
│  >>> s.push(8)                                        │
│  >>> print(s.pop())    # Output: 8                    │
│  >>> print(s.peek())   # Output: 3                    │
│  >>> print(s.pop())    # Output: 3                    │
│  >>> print(s.pop())    # Output: 5                    │
│                                                       │
│  Notice: Elements come out in REVERSE order           │
│  (Last In = 8, First Out = 8)                         │
│                           [OPTIONAL_ENRICHMENT]       │
````

```
└──────────────────────────────────────────────────┘

🖼️ ┌──────────────────────────────────────────────┐
│                                                 │
│ [GENERATED IMAGE: Stack Operations Visualization] │
│                                                 │
│ Mixed style image showing:                      │
│ - Left: Step-by-step push/pop sequence          │
│ - Center: Visual stack growing and shrinking    │
│ - Right: LIFO principle explanation card        │
│ - Bottom: Complexity table                      │
│   push O(1), pop O(1), peek O(1), search O(n)   │
│                                                 │
│ [🔄 Regenerate] [🔄 Different Style ▼]           │
│ [✅ Approve Image]                               │
└─────────────────────────────────────────────────┘

┌─ 📊 Complexity Table ────────────────────────────┐
│                                                  │
│ | Operation | Time   | Space | Notes          | │
│ |-----------|--------|-------|----------------| │
│ | Push      | O(1)   | O(1)  | Always at top  | │
│ | Pop       | O(1)   | O(1)  | Always from top | │
│ | Peek      | O(1)   | O(1)  | Just read top  | │
│ | Search    | O(n)   | O(1)  | Must traverse  | │
│ | isEmpty   | O(1)   | O(1)  | Check length   | │
│                                                  │
└──────────────────────────────────────────────────┘

## Real-World Applications

┌─ 🌍 Application Box ─────────────────────────────┐
│                                                  │
│ Where stacks are used:                           │
│                                                  │
│ 🔙 Browser Back Button                           │
│   Each page visit is pushed to stack             │
│   Back button = pop the stack                    │
│                                                  │
│ ↩️ Undo in Text Editors                          │
│   Each action pushed, undo = pop                 │
│                                                  │
│ 📞 Function Call Stack                           │
│   Each function call is pushed                   │
│   Return = pop and resume previous               │
│                                                  │
│ 🧮 Expression Evaluation                         │
│   Postfix notation uses stack                    │
│                     [OPTIONAL_ENRICHMENT]        │
└──────────────────────────────────────────────────┘

─── NO Current Affairs Section ───────────────────
(Feature was auto-hidden for CS subject)

─── AI Validation same as PSC ────────────────────
[🔍 AI Validate This Part] → same flow
```

Screen 8: Export

What User Sees

```
┌──────────────────────────────────────────────────────┐
│ 🎉 All 6 parts approved!                             │
│                                                      │
│ ── Export Settings ──                                │
│                                                      │
│ Format:                                              │
│ ○ HTML (styled webpage)                              │
│ ○ PDF (via browser print)                            │
│ ○ Markdown (plain text with formatting)              │
│ ○ Copy to clipboard                                  │
│                                                      │
│ Include:                                             │
│ ☑ Cover page (topic name, date, exam)                │
│ ☑ Table of contents                                  │
│ ☑ All generated images (inline)                      │
│ ☑ Current affairs sections                           │
│ ☑ PYQ section                                        │
│ ☑ Source tags (green/blue/gold labels)               │
│ ☐ Blueprint prompts (for reference)                  │
│                                                      │
│ [👁️ Preview] [⬇️ Download]                            │
│                                                      │
│ ── Preview ──                                        │
│                                                      │
│ ┌──────────────────────────────────────────────┐     │
│ │                                              │     │
│ │ ╔══════════════════════════════════════╗     │     │
│ │ ║ NATIONAL INCOME                      ║     │     │
│ │ ║ Comprehensive Study Notes            ║     │     │
│ │ ║ Target: MPSC | State: MP             ║     │     │
│ │ ║ Generated: January 2025              ║     │     │
│ │ ╚══════════════════════════════════════╝     │     │
│ │                                              │     │
│ │ TABLE OF CONTENTS                            │     │
│ │ 1. Foundations of National Income ........... p.1 │
│ │ 2. Types of GDP & Methods .................. p.5 │
│ │ 3. NDP, GNP, NNP & Other Measures ......... p.12 │
│ │ 4. GDP Deflator, Base Year, India .......... p.18 │
│ │ 5. GDP, Welfare & Alternatives ............. p.23 │
│ │ 6. HDI, Indices & PYQs .................... p.28 │
│ │                                              │     │
│ │ [Full rendered preview scrollable here]      │     │
│ │                                              │     │
│ └──────────────────────────────────────────────┘     │
│                                                      │
└──────────────────────────────────────────────────────┘
```

Summary: PSC vs CS — Feature Comparison

Feature	PSC Content	CS Content
Current Affairs	✅ 4 search queries per part (National recent, National major, State recent, State major)	❌ Hidden by default
PYQ Detection	✅ Detect year, type, answer	⚠️ Only if academic exam Qs exist
State Focus	✅ MP/UP specific data	❌ Not applicable
Legal/Case Law	✅ If constitutional/policy topic	❌ Not applicable
Formula Boxes	✅ Economics/Science formulas	✅ Algorithm formulas
Code Blocks	❌ Not applicable	✅ With syntax highlighting
Input/Output Demos	❌ Not applicable	✅ Shows program execution
Complexity Tables	❌ Not applicable	✅ Big-O analysis
Story/Scenario	✅ Character-driven examples	✅ Real-world analogies
Comparison Tables	✅ GDP vs GNP, etc.	✅ Array vs LinkedList, etc.
Timeline	✅ Historical evolution	✅ Technology evolution
Images	✅ Mixed styles (charts + tables + cards)	✅ Mixed styles (diagrams + tables + code visualization)
AI Validation	✅ Same flow	✅ Same flow
AI Chat	✅ Same flow	✅ Same flow
Traceability Tags	✅ Source/Research/Enrichment	✅ Source/Research/Enrichment
Containers	Concept Box, Formula Box, Story Box, Confusion Alert, Legal Box, Timeline Strip	Code Block, I/O Demo, Complexity Table, Application Box, Concept Box

StudyForge Application Skeleton (Unified HTML)

Here is the finalized HTML skeleton with the CSS styles for your special containers explicitly built-in, and the correct 8-screen logic applied.

```
<!DOCTYPE html>
 html lang "en"
 head
   meta charset "UTF-8"
   title StudyForge Skeleton title
   style
   :root
         #1a1a1a
             #2d2d2d
          #f4f0eb
             #4a90e2
             #27ae60
     /* Container Colors */
               #27ae60
              #2980b9
              #f39c12

   body                         var          var          0              100vh
   #sidebar      260px      var                    1px     #444      20px
   #main     1       40px
```

```
      /* Navigation Styles */
      .screen
      .screen.active
      .nav-item              10px                                        4px                   5px           0.9em
      .nav-item:hover                  #444
      .nav-item.active                 var
      button               var                                      10px 20px            4px                              10px

      /* Specialized Content Containers */
      .box-container                 4px      #555          15px         15px 0                rgba 255 255 255 0.05                0 4px 4px 0
      .box-confusion                          #e74c3c   /* Red for confusion alerts */
      .box-legal                         #8e44ad     /* Purple for legal/case law */
      .box-story                         #f39c12     /* Gold for stories/scenarios */
      .box-io                         #16a085                            /* Teal for I/O Demos */
      .box-title                                          8px
     style
   head
 body
     nav id  "sidebar"
         h2 StudyForge  h2
         div id  "nav-links"    div
      nav

     main id  "main"
       <!-- Screen 1: INPUT -->
         section id  "screen-input" class  "screen"
             h1 Step 1: Provide Study Material  h1
             p Upload a file, type a topic, or paste text below:  p
             input type  "file" id  "file-input"   br   br
             textarea placeholder  "Or paste text..." rows  "5" style  "width:100%; background: #333; color: white; border: 1px solid #555; padding: 10px;"    textarea   br
             button onclick  "app.navigate('reading')"  Continue  button
          section

       <!-- Screen 2: READING -->
         section id  "screen-reading" class  "screen"
             h1 Step 2: Reading Content...  h1
             p Processing your inputs. Please wait.  p
             div style  "width: 100%; height: 20px; background: #444; border-radius: 10px; margin: 20px 0;"
                 div style  "width: 65%; height: 100%; background: var(--primary); border-radius: 10px;"    div
              div
             button onclick  "app.navigate('understanding')"  Simulate Done  button
          section

       <!-- Screen 3: UNDERSTANDING -->
         section id  "screen-understanding" class  "screen"
             h1 Step 3: Feature Detection  h1
             div class  "box-container"
                 span class  "box-title"  ✅ Economics Detected  span
                 p 19 topics, 20 definitions, 12 formulas found.  p
              div
             button onclick  "app.navigate('parts')"  Continue to Parts  button
          section

       <!-- Screen 4: PARTS -->
         section id  "screen-parts" class  "screen"
             h1 Step 4: Content Segmentation  h1
             p I will split your material into 6 teachable parts.  p
             button onclick  "app.navigate('blueprint')"  Create Blueprint  button
          section

       <!-- Screen 5: BLUEPRINT (CARD GRID) -->
         section id  "screen-blueprint" class  "screen"
             h1 Step 5: Review Blueprint  h1
             div class  "box-container" style  "cursor: pointer;" onclick  "app.navigate('blueprint_detail')"
                 span class  "box-title"  📝 Part 2: Types of GDP & Measurement Methods  span
                 p Click to open detailed interactive workspace...  p
              div
             button onclick  "app.navigate('generation')"  Approve All & Start Generation  button
          section

       <!-- Screen 6: BLUEPRINT DETAIL VIEW -->
         section id  "screen-blueprint_detail" class  "screen"
             h1 Step 6: Blueprint Detail View  h1
             p Interactive workspace to edit prompts, adjust text plans, and chat with AI.  p
             button onclick  "app.navigate('blueprint')"  ⟵ Back to Cards  button
             button onclick  "app.navigate('generation')"  Approve & Start Generation  button
          section

       <!-- Screen 7: GENERATION -->
         section id  "screen-generation" class  "screen"
             h1 Step 7: Generating Notes  h1
             p Executing blueprints...  p

           <!-- Example of stylized containers rendering in standard flow -->
             div class  "box-container box-confusion"
                 span class  "box-title"  ⚠️ Common Confusion Alert  span
                Nominal GDP growing does NOT mean the economy is growing.
              div

             div class  "box-container box-story"
                 span class  "box-title"  📖 Story Box: Shyam The Bread Maker  span
                Value Added = Value of Output − Intermediate Consumption
              div

             div class  "box-container box-io"
                 span class  "box-title"  🔄 Input ⟶ Output Demo  span
                >>> s = Stack()  br
                >>> s.push(5)  br
                >>> print(s.pop()) # Output: 5
              div

             button onclick  "app.navigate('export')"  Approve All & Continue to Export  button
          section

       <!-- Screen 8: EXPORT -->
         section id  "screen-export" class  "screen"
             h1 Step 8: Final Document  h1
             p Your notes are ready to download.  p
             button onclick  "alert('Downloading HTML/PDF...')"  Download Final Output  button
          section
      main

     script
     const
        // Correctly tracking all 8 logical screens
        state   currentScreen  'input'  data  null  parts     approved
        screens   'input'  'reading'  'understanding'  'parts'  'blueprint'  'blueprint_detail'  'generation'  'export'

        init
          this
          this            this


        navigate
          this
          document                     '.screen'                                  'active'
          document                     `screen-${screenId}`                'active'
          this
```

```
    renderNav
      const          document                          'nav-links'
                          this                                         `
        <div class="nav-item ${this.state.currentScreen === s ? 'active' : ''}" onclick="app.navigate('${s}')">
          ${index + 1}. ${s.replace('_', ' ').toUpperCase()}
        </div>`          ''


  script
body
html
```